

Math for CS & AI

Bobby

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Discrete Probability

Problem 1

Consider n aliens living on a planet with m days in a year. Let A be the event that at least two aliens share the same birthday.

Prove that the collision probability $\Pr[A]$ satisfies:

$$\Pr[A] = 1 - e^{-N/m} + \mathcal{O}\left(\frac{1}{\sqrt{m}}\right)$$

as $m \rightarrow \infty$, where $N := n(n-1)/2$.

Hint 1

It is much easier to first calculate the probability of no collisions, denoted as $\bar{A}$.

$$\Pr[A] = 1 - \Pr[\bar{A}] = 1 - \prod_{k=0}^{n-1} \left(1 - \frac{k}{m}\right)$$

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Hint 2

The asymptotic behavior of the error term depends heavily on the growth rate of n . Decompose the proof into two cases based on a threshold:

- Case 1: $n \leq m^{0.6}$ (Small n)
- Case 2: $n > m^{0.6}$ (Large n)

Hint 3

Take the natural logarithm of $\Pr[\bar{A}]$ and apply the Taylor expansion $\ln(1 - x) = -x + \mathcal{O}(x^2)$:

$$\ln \Pr[\bar{A}] = \sum_{k=0}^{n-1} \ln \left(1 - \frac{k}{m} \right) = \underbrace{-\sum_{k=0}^{n-1} \frac{k}{m}}_{\text{Main term } -N/m} + \underbrace{\mathcal{O} \left(\sum_{k=0}^{n-1} \frac{k^2}{m^2} \right)}_{\text{Error term } \mathcal{O}(n^3/m^2)}$$

Hint 4

Use the standard inequality $1 - x \leq e^{-x}$ for all $x \in \mathbb{R}$ to upper bound the product directly:

$$\Pr[\bar{A}] \leq \prod_{k=0}^{n-1} e^{-k/m} = e^{-N/m}$$

Then, show that for large n , both $\Pr[\bar{A}]$ and $e^{-N/m}$ become negligibly small and are absorbed by the $\mathcal{O}\left(\frac{1}{\sqrt{m}}\right)$ error bound.

Problem 2

Consider throwing n balls independently and uniformly at random into n bins. Let M be the maximum load (the maximum number of balls in any single bin). Prove that with high probability, the maximum load M satisfies:

$$M = \mathcal{O}\left(\frac{\ln n}{\ln \ln n}\right)$$

Specifically, show that for a sufficiently large constant c , $\Pr\left[M \geq \frac{c \ln n}{\ln \ln n}\right] \leq \frac{1}{n}$.

Hint 1

Focus on a single bin, say Bin i . Let X_i be the number of balls in Bin i . This follows a Binomial distribution $B(n, 1/n)$. The probability that Bin i contains at least k balls is:

$$\Pr[X_i \geq k] \leq \binom{n}{k} \left(\frac{1}{n}\right)^k$$

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Hint 2

Use Stirling's approximation or the simpler inequality $\binom{n}{k} \leq \left(\frac{ne}{k}\right)^k$ to simplify the bound:

$$\Pr[X_i \geq k] \leq \frac{n^k}{k!} \cdot \frac{1}{n^k} = \frac{1}{k!} \leq \left(\frac{e}{k}\right)^k$$

Hint 3

Apply the Union Bound over all n bins to bound the maximum load M :

$$\Pr[M \geq k] = \Pr\left[\bigcup_{i=1}^n \{X_i \geq k\}\right] \leq \sum_{i=1}^n \Pr[X_i \geq k] \leq n \left(\frac{e}{k}\right)^k$$

We want this probability to be $\leq n^{-1}$, which means we need $n(e/k)^k \leq n^{-1}$, or equivalently $k(\ln k - 1) \geq 2 \ln n$.

Hint 4

To solve $k \ln k \approx \ln n$, let $k = \frac{c \ln n}{\ln \ln n}$. Substitute this into the inequality:

$$\ln k = \ln \left(\frac{c \ln n}{\ln \ln n} \right) = \ln \ln n - \ln \ln \ln n + \ln c$$

Show that for a large enough c , the $k \ln k$ term dominates $\ln n$ and satisfies the bound. This result highlights the "power of choices" in randomized algorithms.

Problem 3

Let $G = (V, E)$ be a graph with n vertices and m edges, where $n \leq 2m$. Use the probabilistic method to prove that there exists an independent set in G of size at least:

$$\alpha(G) \geq \frac{n^2}{4m}$$

where $\alpha(G)$ denotes the size of the maximum independent set.

Hint 1

Construct a random subset $S \subseteq V$ by independently including each vertex $v \in V$ with probability p . Let X be the number of vertices in S , and Y be the number of edges in the subgraph induced by S . Show that:

$$\mathbb{E}[X] = np, \quad \mathbb{E}[Y] = mp^2$$

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Hint 2

S is not necessarily an independent set. To obtain an independent set S' , for each edge in the induced subgraph, remove one of its endpoints from S . The size of the resulting independent set S' satisfies:

$$|S'| \geq X - Y$$

Apply the linearity of expectation to find $\mathbb{E}[|S'|]$.

Hint 3

We have $\mathbb{E}[|S'|] \geq np - mp^2$. To find the tightest bound, maximize this quadratic function with respect to p :

$$f(p) = np - mp^2 \implies f'(p) = n - 2mp = 0$$

Set $p = \frac{n}{2m}$ (Note: $n \leq 2m$ ensures $p \leq 1$). Substitute p back into the expectation to obtain the final bound $\frac{n^2}{4m}$.

Hint 4

By the property of expectation, if the average value of $|S'|$ is $\frac{n^2}{4m}$, then there must exist at least one specific set S' in the sample space such that:

$$|S'| \geq \frac{n^2}{4m}$$

This completes the proof. This method is a powerful example of using **alteration** to satisfy hard constraints in combinatorics.

Sampling and Concentration

Problem 1

For any $k \geq 3$, prove that there exists a graph G such that its chromatic number $\chi(G) > k$ and its girth $g(G) > k$ (where $g(G)$ is the length of the shortest cycle in G).

Hint 1

Fix $\epsilon < 1/k$. Consider a random graph $G(n, p)$ with $p = n^{\epsilon-1}$. Let X be the number of cycles in G of length at most k . Show that the expected number of such short cycles is:

$$\mathbb{E}[X] = \sum_{j=3}^k \frac{(n)_j}{2j} p^j = \mathcal{O}(n^k p^k) = \mathcal{O}(n^{k\epsilon})$$

Since $k\epsilon < 1$, $\mathbb{E}[X] = o(n)$. By Markov's inequality, $\Pr[X \geq n/2] \rightarrow 0$ as $n \rightarrow \infty$.

Hint 2

Recall that $\chi(G) \geq n/\alpha(G)$. We need to upper bound the independence number $\alpha(G)$. Show that for $r = \lceil \frac{3}{p} \ln n \rceil$:

$$\Pr[\alpha(G) \geq r] \leq \binom{n}{r} (1-p)^{\binom{r}{2}} \leq \left[\frac{ne}{r} e^{-p(r-1)/2} \right]^r$$

Prove that this probability tends to 0 as $n \rightarrow \infty$. Thus, w.h.p., $\alpha(G) < \frac{3}{p} \ln n = 3n^{1-\epsilon} \ln n$.

Hint 3

Choose n large enough such that $\Pr[X \geq n/2] < 1/2$ and $\Pr[\alpha(G) \geq n/2k] < 1/2$. There exists a graph G with:

- Fewer than $n/2$ cycles of length $\leq k$.
- $\alpha(G) < n/2k$.

To obtain G' , remove one vertex from each cycle of length $\leq k$. The resulting graph G' has $|V'| \geq n/2$ vertices and **no cycles of length $\leq k$** .

Hint 4

For the modified graph G' , the girth $g(G') > k$ by construction. The independence number $\alpha(G')$ cannot increase by removing vertices, so $\alpha(G') \leq \alpha(G) < n/2k$. Then, the chromatic number of G' satisfies:

$$\chi(G') \geq \frac{|V'|}{\alpha(G')} \geq \frac{n/2}{n/2k} = k$$

With a slightly more careful choice of constants, we ensure $\chi(G') > k$.

Problem 2

In the Coupon Collector's Problem, let T be the number of trials to collect all n distinct coupons. 1. Find the variance $\text{Var}(T)$. 2. Prove that as $n \rightarrow \infty$, the variance satisfies the upper bound:

$$\text{Var}(T) \leq \frac{\pi^2}{6} n^2$$

Hint 1

Let X_i be the number of trials to collect the i -th new coupon after $i - 1$ distinct coupons have been collected. X_i follows a Geometric distribution $G(p_i)$ with success probability:

$$p_i = \frac{n - (i - 1)}{n}$$

Since X_i are independent, the total time is $T = \sum_{i=1}^n X_i$, and the variance is:

$$\text{Var}(T) = \sum_{i=1}^n \text{Var}(X_i) = \sum_{i=1}^n \frac{1 - p_i}{p_i^2}$$

Hint 2

Substitute p_i into the variance formula:

$$\text{Var}(T) = \sum_{i=1}^n \frac{1 - \frac{n-i+1}{n}}{\left(\frac{n-i+1}{n}\right)^2} = \sum_{i=1}^n \frac{n(i-1)}{(n-i+1)^2}$$

Let $j = n - i + 1$. As i ranges from 1 to n , j ranges from n down to 1:

$$\text{Var}(T) = \sum_{j=1}^n \frac{n(n-j)}{j^2} = n^2 \sum_{j=1}^n \frac{1}{j^2} - n \sum_{j=1}^n \frac{1}{j}$$

Hint 3

To obtain the upper bound, we drop the negative term and consider the limit of the Basel series:

$$\text{Var}(T) < n^2 \sum_{j=1}^n \frac{1}{j^2} < n^2 \sum_{j=1}^{\infty} \frac{1}{j^2}$$

Using the Euler's solution to the Basel problem, we know:

$$\sum_{j=1}^{\infty} \frac{1}{j^2} = \frac{\pi^2}{6}$$

Hint 4

Therefore, we have the tight asymptotic bound:

$$\text{Var}(T) \leq \frac{\pi^2}{6} n^2$$

This result shows that while the expected time $\mathbb{E}[T] \approx n \ln n$, the standard deviation $\sigma(T)$ is of order $\Theta(n)$, meaning the distribution is relatively concentrated for large n .